1 Introduction

The question of whether a closed manifold admits a metric of positive scalar curvature has been a central topic in differential geometry over the past decades and remains an area of active research. One half of obtaining an answer to this question consists in finding suitable obstructions to the existence of positive scalar curvature metrics. For example in dimension 2 the Euler characteristic of a closed manifold is such an obstruction by the Gauß-Bonnet Theorem and by the classification of surfaces this is in fact the only obstruction. However in higher dimensions this question becomes much more subtle. One way to still approach it is by means of index theory of the spin Dirac operator. In 1980 Gromov and Lawson [**?**] have used this technique to identify a large class of manifolds, which do not admit positive scalar curvature metrics, namely so called enlargeable manifolds. We will give a precise definition of this notion below, from which we will easily derive the following two facts:

1. S^1 is enlargeable,
2. Products of enlargeable manifolds are enlargeable.

From this it follows at once that the n-Torus T^n is enlargeable and thus does not admit a metric of positive scalar curvature for all dimensions $n \geq 0$, which had long been an open question. This example already highlights the importance of the concept of enlargeabilty, which, since its emergence, has been a subject of extensive research. More generally, the two facts above show that if M is enlargeable, $M \times S^1$ does not admit a metric of positive scalar curvature. There are two natural ways one may try to generalize this statement. Firstly, we can ask what happens if we replace M by any non-enlargeable manifold, which does not admit a metric of positive scalar curvature. Naively one might expect the statement to still be true, nonetheless it turns out that there are counterexamples, see [**?**, Remark 1.4], [**?**, Counterexample 4.16]. Secondly, we can ask what happens if the S^1-factor appears as the fiber of some non-trivial circle bundle over M. However, once again, this seemingly plausible expectation does not hold. Counterexamples to this second question have recently been constructed in [**?**] by relying on techniques from symplectic geometry. In the following we will provide an alternative construction for such counterexamples, employing only basic topological notions. Let us begin by recalling the definition of enlargeabilty.

Definition 1.1. A closed oriented Riemannian n-manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a finite Riemannian covering $\hat{M} \to M$ together with an ε-contracting (that is ε-Lipschitz) map $\hat{M} \to S^n$ of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M, since for closed manifolds all metrics are in bi-Lipschitz correspondence. Our definition of enlargeabilty is adopted from [**?**] and [**?**] in contrast to the orginial definition in [**?**], where the coverings $\hat{M}$ were additionally required to be spin. The observation that enlargeabilty is an obstruction to the existence of positive scalar curvature metrics was first shown by Gromov and Lawson [**?**] in the case where M is spin. This was later generalized by Cecchini and Schick [**?**] to the non-spin case.

As already mentioned above, the simplest example for an enlargeable manifold is S^1. To see this consider the Riemannian covering $(S^1, ng) \to (S^1, g), z \mapsto z^n$, where g is the standard metric on S^1. Then the map $(S^1, ng) \to (S^1, g), z \mapsto z$ is ${}^1\!/\!{}_n$-Lipschitz and of degree 1. Since we can choose n arbitrarily large it follows that S^1 is enlargeable.

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 1.2. *(i) The product of enlargeable manifolds is enlargeable.*

(ii) A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.

Proof. For (i) let M^m and N^n be enlargeable. Given $\varepsilon > 0$ choose Riemannian coverings $p\colon \hat{M} \to M$ and $q\colon \hat{N} \to N$ as well as ε-contracting maps $f\colon \hat{M} \to S^m$ and $g\colon \hat{N} \to S^n$ of non-zero degree. Then

$$p \times q\colon \hat{M} \times \hat{N} \to M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover

$$f \times g\colon \hat{M} \times \hat{N} \to S^m \times S^n$$

defines an ε-contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi\colon S^m \times S^n \to S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := {S^m \times S^n}\big/(S^m \times * \cup * \times S^n),$$

which is well known to be homeomorphic to the sphere S^{m+n}. The map ϕ is easily seen to be of degree ± 1 (depending on the chosen orientations) and after a slight perturbation we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that ϕ is a λ-Lipschitz map for some constant $\lambda > 0$. Therefore $\phi \circ (f \times g)$ is a $\lambda\varepsilon$-Lipschitz map of non-zero degree and since ε was chosen arbitrarily the statement follows.

For (ii) let M^m be enlargeable and let $h\colon P \to M$ be some non-zero degree map. Again given $\varepsilon > 0$ choose a Riemannian covering $p\colon \hat{M} \to M$ as well as an ε-contracting map $f\colon \hat{M} \to S^m$. Define a covering $\pi\colon \hat{P} \to P$ as the pullback covering of p under h, that is π fits into a pullback diagram of the form

$$\begin{array}{ccc} \hat{P} & \xrightarrow{H} & \hat{M} \\ \big\downarrow{\scriptstyle \pi} & & \big\downarrow{\scriptstyle p} \\ P & \xrightarrow{h} & M \end{array}$$

where H is some suitable smooth map. Since $\hat{M}$ and P are closed, the same holds for $\hat{P}$. Moreover h is μ-Lipschitz for some constant $\mu > 0$, because P is closed. Consequently H is also μ-Lipschitz, given that both π and p are local isometries. For the degree of H we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(p) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that H is of non-zero degree, such that $f \circ H\colon \hat{P} \to S^m$ is a $\mu\varepsilon$-Lipschitz map of non-zero degree. Again since ε was chosen arbitrarily the statement follows. $\square$

From Part (ii) of the above Lemma it follows immediately, that enlargeabilty is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1. In fact more is true, ...

2 The Construction

We will now construct counterexamples to the question raised in the beginning, that is circle bundles $E \to M$ over enlargeable M, where the total space E admits a metric of positive scalar curvature. In the following we will refer to orientable manifolds with non-spin universal cover as *totally non-spin*.

For $n \geq 2$ let N^{2n} be any $2n$-dimensional totally non-spin manifold. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma A.1 and Lemma A.3 (see Appendix). Now consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \to T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 1.2 that M is also enlargeable. We will now construct a circle bundle $E \to M$ as follows. Let

$$c_1 \in H^2(M;\mathbb{Z}) \cong H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n};\mathbb{Z}) \oplus H^2(N;\mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n;\mathbb{Z})$ of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$. We then define $E \to M$ as the unique S^1-principal bundle with first chern class equal to c_1. Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that E is trivial over the summand $T^{2n} \# N$ and is given by the restriction of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1 \colon (T^{2n} \# N) - D^{2n} \to M$$

be the inclusion. Then

$$\begin{tikzcd}
H^2(M;\mathbb{Z}) \arrow[r, "i_1^*"] & H^2((T^{2n}\#N) - D^{2n};\mathbb{Z}) \\
H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"]
\end{tikzcd}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that S^1-principal bundles are uniquely determined by their first chern class we see that i_1^*E is indeed trivial. Similiarly if we denote by

$$i_2 \colon \mathbb{C}P^n - D^{2n} \to M, \quad j \colon \mathbb{C}P^n - D^{2n} \to \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{tikzcd}
H^2(M;\mathbb{Z}) \arrow[r, "i_2^*"] & H^2(\mathbb{C}P^n - D^{2n};\mathbb{Z}) \\
H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(\mathbb{C}P^n;\mathbb{Z}) \arrow[u, "\cong"', "j^*"]
\end{tikzcd}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of i_2^*E and j^*S^{2n+1} agree, hence they must be isomorphic bundles.

With this intuition in mind we will now show the following two claims.

Claim 1. E is totally non-spin.

Claim 2. The projection $p \colon E \to M$ induces an isomorphism on π_1.

Granting these claims for now, let us see how to conclude the existence of a positive scalar curvature metric on E. The disk bundle $DE \to M$ corresponding to E defines an oriented null-cobordism of E. Moreover, since $DE \to M$ is a homotopy equivalence it follows from Claim 1 that the inclusion $E \to DE$ is an isomorphism on π_1. So together with Claim 2 the following Theorem applies.

Theorem 2.1. *Let X^n be a closed connected totally non-spin manifold of dimension $n \geq 5$. Suppose there exists an oriented null-cobordism W of X, such that the inclusion $X \to W$ induces an isomorphism on π_1. Then X admits a metric of positive scalar curvature.*

We will give a proof of this theorem further below, which is essentially an application of the well known Gromov-Lawson surgery principle (SOURCE). Let us now proof Claim 1 and Claim 2.

Proof of Claim 1. Let $i\colon N - D^{2n} \to M$ be the inclusion. We have already established above, that E is trivial over the $T^{2n}\#N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma A.1 we conclude that $N_0 := N - D^{2n}$ and consequently also $(N - D^{2n}) \times S^1$ are totally non-spin. Therefore E contains a totally non-spin submanifold of codimension 0 and so again Lemma A.1 implies that E is totally non-spin. $\square$

Proof of Claim 2. Let $j\colon \mathbb{C}P^n - D^{2n}$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{tikzcd}
\pi_2(M) \arrow[r, "\partial_1"] & \pi_1(S^1) \arrow[r] & \pi_1(E) \\
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[u] \arrow[d, "\cong"'] \arrow[r, "\partial_2"] & \pi_1(S^1) \arrow[u, "\mathrm{id}"'] \arrow[d, "\mathrm{id}"'] \arrow[r] & \pi_1(j^*E) \arrow[u] \arrow[d] \\
\pi_2(\mathbb{C}P^n) \arrow[r, "\partial_3"] & \pi_1(S^1) \arrow[r] & \pi_1(S^{2n+1})
\end{tikzcd}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{tikzcd}
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\mathbb{C}P^n) \arrow[d, "\cong"] \\
H_2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) \arrow[r, "\cong"] & H_2(\mathbb{C}P^n; \mathbb{Z})
\end{tikzcd}$$

commutes and the vertical maps are isomorphisms. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2, which in turn is equivalent to surjectivity of ∂_3. But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \to \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \to \pi_1(E) \to \pi_1(M) \to \pi_0(S^1),$$

that $\pi_1(E) \to \pi_1(M)$ is indeed an isomorphism. $\square$

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds in all even dimensions ≥ 4. To also get examples in odd dimensions let $E \to M$ be as above and let $\pi\colon M \times S^1 \to M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 1.2 the base space $M \times S^1$ is still enlargeable.

A Appendix

The following lemma collects a few basic facts about totally non-spin manifolds.

Lemma A.1.

(i) The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.

(ii) A manifold containing a totally non-spin submanifold of codimension 0 *is totally non-spin.*

(iii) If M is totally non-spin and of dimension $n \geq 4$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \operatorname{int}(M)$.

Remark A.2. All statements of the lemma remain valid if "totally non-spin" is replaced by "non-spin". The corresponding proofs are just simpler versions of the arguments given below.

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_1, \quad \pi_2 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_2$$

the projections and let $\iota \colon \tilde{M}_1 \to \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned}\iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 + (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0,\end{aligned}$$

where we used that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M. Then the inclusion $\iota \colon N \to M$ lifts to the universal covers:

$$\begin{tikzcd} \tilde{N} \arrow[r, "\tilde{\iota}"] \arrow[d] & \tilde{M} \arrow[d] \\ N \arrow[r, "\iota"] & M \end{tikzcd}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota} \colon T\tilde{N} \to T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{tikzcd} T\tilde{N} \arrow[r, "d\tilde{\iota}"] \arrow[d] & T\tilde{M} \arrow[d] \\ \tilde{N} \arrow[r, "\tilde{\iota}"] & \tilde{M} \end{tikzcd}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without of loss generality that M is connected. Denote by $B^n := \operatorname{int}(D^n)$ the open ball. Then inclusion

$$M - D^n \to M - B^n$$

is a homotopy equivalence from which we conclude (using analogous arguments as for (ii)) that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. We will now show this for the latter. Let $p \colon \tilde{M} \to M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p. From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber f is finite we can apply the Seifer-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi\colon S^1 \to \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi\colon D^2 \to \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic $\operatorname{rel} S^1$ to some map $\Phi_1\colon D^2 \to \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq \tilde{M}$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. Since $n \geq 3$ this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0 = \tilde{M} - p^{-1}(B^n)$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r\colon \tilde{M} - p^{-1}(0) \to \hat{M}_0$$

be such a strong deformation retraction and let

$$r_t\colon \tilde{M} - p^{-1}(0) \to \tilde{M} - p^{-1}(0)$$

be a homotopy from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy $\operatorname{rel} S^1$ from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\tilde{M}_0 = \hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F)$$

Finally the Mayer-Vietoris sequence of the pushout

$$\begin{tikzcd} S^{n-1} \times F \arrow[r] \arrow[d] & D^n \times F \arrow[d] \\ \tilde{M} - (B^n \times F) \arrow[r, "\iota"] & \tilde{M} \end{tikzcd}$$

where all maps are inclusions, then shows that ι is an isomorphism on $H^2(-;\mathbb{Z}_2)$. Here we need $n \geq 4$. Therefore

$$w_2(T\tilde{M}_0) = \iota^* w_2(T\tilde{M}) \neq 0.$$

□

In the construction in section 2 we also made use of the following fact.

Lemma A.3. *$\mathbb{C}P^2$ is totally non-spin.*